

Abdelrahman Mohamed Ismail

AI Engineer

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Summary

AI Engineer specializing in machine learning, deep learning, NLP, forecasting, and RAG systems. Experienced in building end-to-end AI products from data pipelines, model training, evaluation, and retrieval systems to production ready APIs and full-stack applications. Strong background in financial forecasting, LLM based applications, backend engineering, and real world AI system deployment.

Education

Egypt-Japan University of Science and Technology (E-JUST)

Alexandria, Egypt

B.Sc. in Computer Science and Information Technology — AI & Data Science

June 2026

Experience

Orange Egypt — *Data Analytics Intern, Data & Digital Transformation*

Summer 2026

Smart Village, Giza, Egypt

- Worked within the Data & Digital Transformation department on enterprise data analytics, reporting workflows, and operational data analysis for a large telecom operator.
- Analyzed business and customer-related datasets to support internal reporting, dashboarding, and data-driven decision-making workflows across enterprise teams.

Projects

Stock Market Prediction Platform — *Machine Learning, Deep Learning, NLP, RAG*

Graduation Project

Tech Stack: Python, PyTorch, Scikit-learn, FastAPI, Transformers, FinBERT, Qdrant, Docker

- Developed an end-to-end stock market forecasting platform for S&P 500 and Nasdaq-100 equities using machine learning, deep learning, and financial news sentiment analysis.
- Implemented and benchmarked 10+ forecasting architectures, including ElasticNet, Ridge, HistGBR, LSTM, GRU, N-BEATS, N-HiTS, TFT, Informer, PatchTST, Autoformer, FEDformer, and Mamba, using leakage-free walk-forward validation.
- Achieved up to 73.0% directional accuracy on 5-day return forecasting and 65.1% directional accuracy on 1-day forecasting tasks.
- Trained, evaluated, and managed a registry of 2,179 forecasting models spanning classical ML, recurrent neural networks, transformer architectures, and state-space models.
- Built a production-ready RAG explanation system using Qdrant, E5 embeddings, hybrid retrieval, and LLM-generated citation-grounded financial explanations.

Zayed Heights Compound App — *Full-Stack Mobile Application*

Tech Stack: Flutter, Dart, Supabase, PostgreSQL, Edge Functions, FCM

- Built a cross-platform smart residential compound management app with role-based dashboards for residents, security guards, and administrators.
- Implemented QR-based visitor access, PostgreSQL RPCs, Supabase Edge Functions, and real-time gate logging using Supabase Realtime.
- Integrated FCM push notifications, device binding security, audit logging, and role permission access control for secure multi role workflows.

Egyptian Constitution Chatbot — *RAG, Semantic Search*

Tech Stack: LlamaIndex, ChromaDB, Hugging Face, Streamlit, FastAPI

- Built a RAG based chatbot for legal document retrieval and constitutional Q&A using semantic search, vector indexing, and a real time Streamlit interface.

Arabic Text-to-SQL System — *LLM Applications, NLP*

Tech Stack: Gemini 1.5 Flash, Python, FastAPI, ChromaDB

- Developed an Arabic natural language to SQL system with schema aware prompting and RAG retrieval, optimized for low latency query generation across multi table schemas.

Certifications & Training

- **Digital Egypt Pioneers Initiative (DEPI) — Generative AI Professional Track, 200+ Hours:** Built and deployed LLM applications, RAG systems, NLP pipelines, and generative AI workflows.
- **National Telecommunication Institute (NTI) — Data Science & Machine Learning, 120+ Hours:** Developed machine learning and deep learning solutions using TensorFlow, OpenCV, and Python.

Technical Skills

- **Languages:** Python, C++, C#, JavaScript/TypeScript, Dart, SQL
- **ML & Deep Learning:** PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas
- **LLMs, NLP & CV:** RAG, LangChain, LlamaIndex, Hugging Face Transformers, FinBERT, OpenCV, MediaPipe, OCR
- **Backend, Frontend & Databases:** FastAPI, Flask, REST APIs, Streamlit, Flutter, React, PostgreSQL, MySQL, MongoDB, Qdrant, ChromaDB, Supabase
- **Cloud & DevOps:** Docker, GCP, Linux, GitHub Actions, Git, Postman, VS Code